Advice on giving good research talks

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github.com/ChrisHIV/teaching :
these slides + prose version,
previous advice about writing.

Small italic text appearing in boxes contains things I meant to say verbally, for the benefit of those not present in the talk.

Introductory generalities

Writing the talk

Logistical preparation

Delivering the talk

Scope

Normal seminars:

- one paper / a few related ones
- your own work
- to other specialists

Differences apply for:

- talks to public, policy makers
- job/fellowship/grant talks
- subject overview talks
- talks about advice…

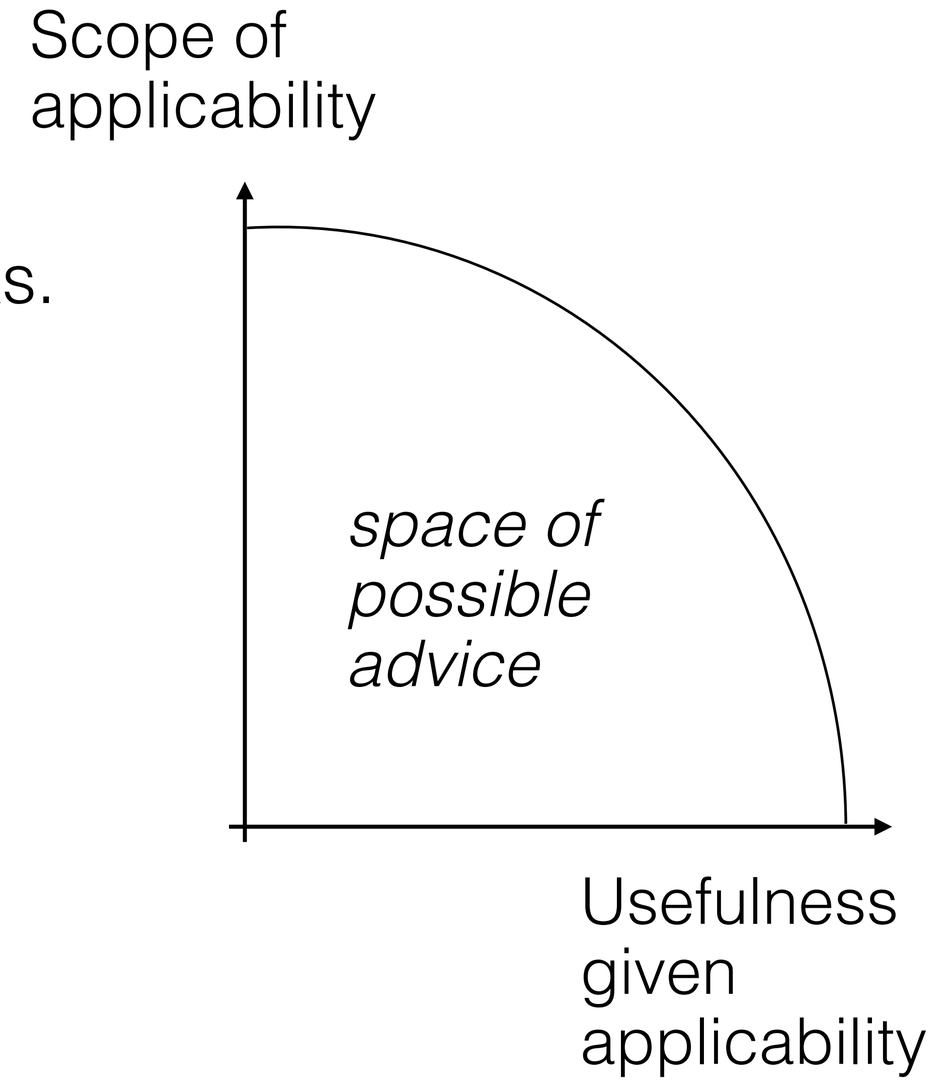
Caveats

There is more than one good way of giving talks.

Knowingly breaking ‘rules’
is better than not knowing any.

Reading critically improves your writing;
listening critically improves your speaking.

Some recommendations cost your time.



Write telegraphically

HIV

The causative pathogen for HIV is an RNA virus.

As of 2025, an estimated 41 million individuals live with HIV. In 2025 there were an estimated 1.1 million new infections.

HIV is most commonly transmitted sexually, but is also transmitted through the sharing of contaminated needles.

→

HIV

RNA virus

Prevalence: 41 million.
Incidence: 1.1 million / year.

Transmission: mostly sexual, also needles

Things I said verbally: see next slide

Reduce reading time, increase listening time.

Writing supports listening before, during and after.

Some tradeoff with slides' standalone usefulness, but not much.

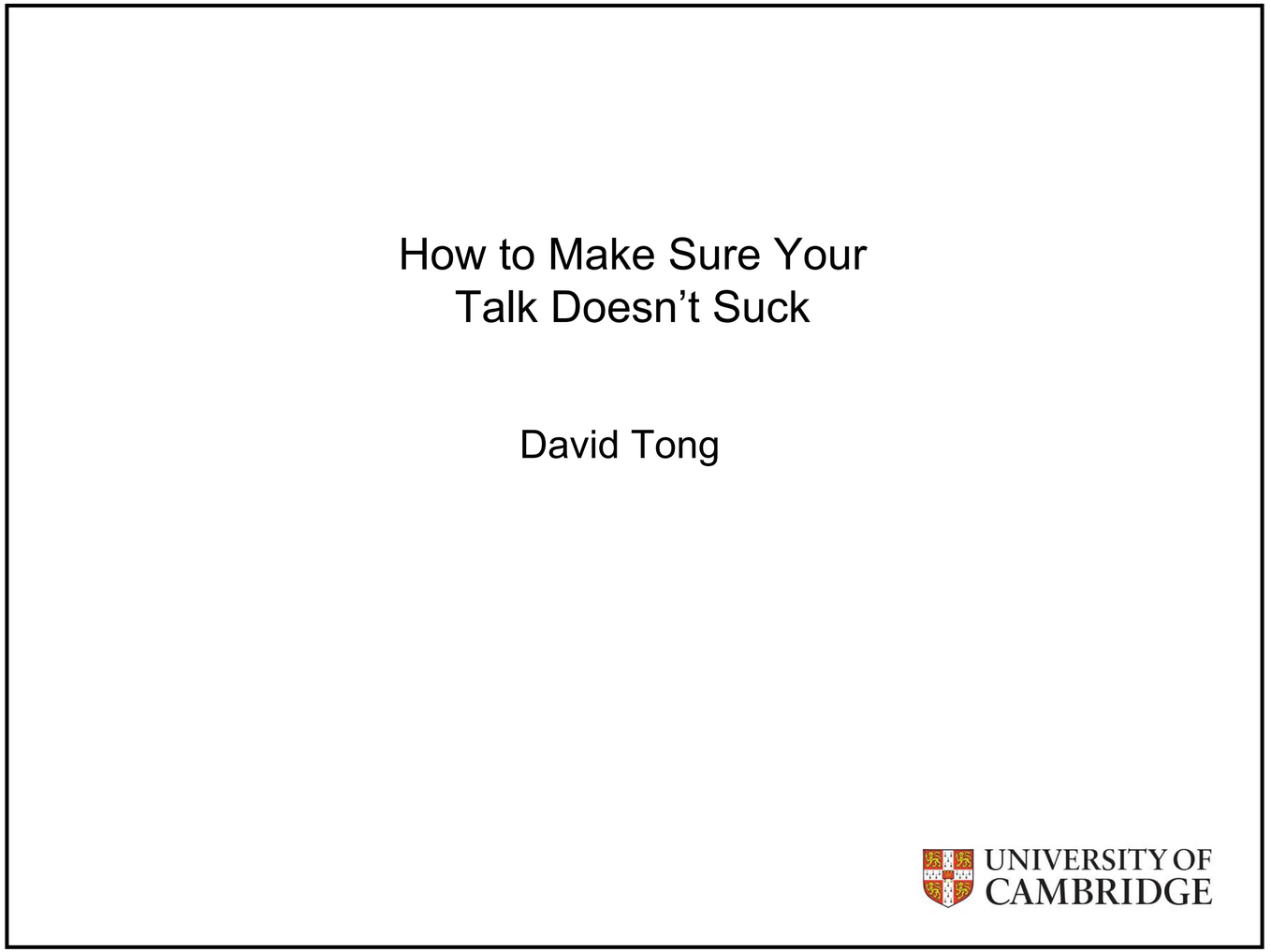
Take a look at these two slides.
…
I've brought this bit into the front section of my talk, both because it's important, and because after hearing this piece of advice, it might help to see me following myself it in the rest of the talk.
I only follow it sometimes today: it's harder advice to follow for an advice talk than a research talk.
The advice is: write telegraphically.
You don't need to write in complete grammatically correct sentences: the text is just a prop for what you'll say out loud.
Try to reduce everything to the minimum number of words required for the listener to know what you mean.
Why is this helpful?
You want people to listen to you, that's why you're talking.
The less time they need to read, the more they can listen to you.
The writing serves three main supporting roles - before, during and after listening to you.
Before you start talking, some listeners will first read the whole slide to get the big picture of what you're about to say, so they can then use your spoken words to fill out the details.
While you're talking, if someone gets lost, they can look at the text again to get unlost.
After you've finished talking through a slide, listeners can read again to recap what they just learned.
For all three of those purposes it helps if they can read quickly, because you're not going to be giving them long periods of silence for reading.
The full-sentence version of the content comes in what you say out loud.
I would typically say things like the left slide but write things like on the right slide.
There is some tradeoff between serving people who actually attend your talk and serving people who only see your slides afterwards.
However, first, you can minimise the tradeoff by writing telegraphically skilfully - pick out the key words such that someone reading knows what you mean.
Second, I think your primary responsibility is serving the people who actually attend.
Third, it's an option to add content to your slides afterwards or have a document form of the content.
Fourth, this is only a rule of thumb. If you have one specific talk where you know an important stakeholder won't be there, you know they'll want the slides, and you only have time to write the slides once, for that talk be less telegraphic, fine.

Write legibly, consistently.

Speak audibly, consistently.

I've pulled this slide out to the front section of the talk, both because these points are important, and to give an early example of the principle that less is more.
When I only write two short things on the slide, they each get a lot of emphasis, and I can be more confident that you will come away knowing what were the two things I wanted to say.
Enough about the style of the slide, now for the content.
Write legibly, speak audibly, and do so everywhere throughout your entire presentation.
People write smaller for things that are less important; to an extent that's OK, but every bit of text needs to be big enough to be easy to read.
People speak more quietly when they're nervous; that's understandable, but try to find a way not to punish people who are straining to hear you.
People also start speaking more quietly when they're in the middle of saying something and they realise it's not important, or they're uncertain of what they're saying: they trail off quietly into a mumble.
Decide whether something is worth saying or not, if it is, everyone should be able to hear it, if it isn't, don't say it.
If you change your mind mid-sentence, don't trail off into a mumble, just say, "actually forget about it" and move on.
Don't use the boundary between what's easy to read and what's not, what's easy to hear and what's not, as the place for your less important points.
Always stay on the good side of the boundary.
For speaking, when you're addressing a whole room of people, you need to increase the volume of your voice compared to a normal conversation.
(Or use a microphone.)

Inspiration



davidtong.org/pdfs/essays/talk.pdf

Introductory generalities

Writing the talk

Logistical preparation

Delivering the talk

Recapping relevant writing advice, i

Intro: general background → specific background → clear, motivated question

Materials & Methods section: order steps chronologically.

Follow Results by implications/outlook: so what, what next?

I've previously discussed advice for writing in greater detail (see github.com/ChrisHIV/teaching); here I'll briefly recap those points that also apply to giving talks.

Recapping relevant writing advice, ii

Choose words for clarity, not impressiveness.
Make it broadly accessible.

Each point = information or persuasion. Be clear which.

Be specific about your work's value.
Don't just say you *characterised* some system.

Use acronyms sparingly. Define at first use.

Recapping relevant writing advice, iii

Crafting each sentence (e.g. in a script):

- start with what's familiar, simple, expected
- align [main actors, actions] with [grammatical subjects, verbs]

One slide ≈ one paragraph:

- one main point per slide
- plan and analyse at the slide level
- variation in amount of content is OK (but never too much)

Key questions

How long, excluding questions?

For whom?

What are your 1-4 main takeaways?
Build up to those,
not down from *everything*.
Less, slowly > more, quickly.

Ordering sections

Materials & Methods before Results before Discussion

As whole sections, or by sub-analysis

Show structure

For longer talks. e.g. slides like

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Limit text

No more than ~5 points per slide

Some text usually better than none.
None works for a photo + story,
or a plot with one clear message.

Limit tabulated content

area	x	y
A	5.37	9.47
B	5.83	10.66
C	1.35	11.59
D	4.26	10.56
E	3.67	11.03
F	3.94	9.31
G	4.43	10.33
H	4.14	10.95
I	3.98	11.23
J	4.53	8.77
K	1.98	9.65
L	4.54	9.18



area	x	y
J	4.53	8.77
K	1.98	9.65

Don't display a big table of values, most of which you want the listeners to ignore. If only part of a table is relevant, extract that part.

Timing

Formal talks: time yourself

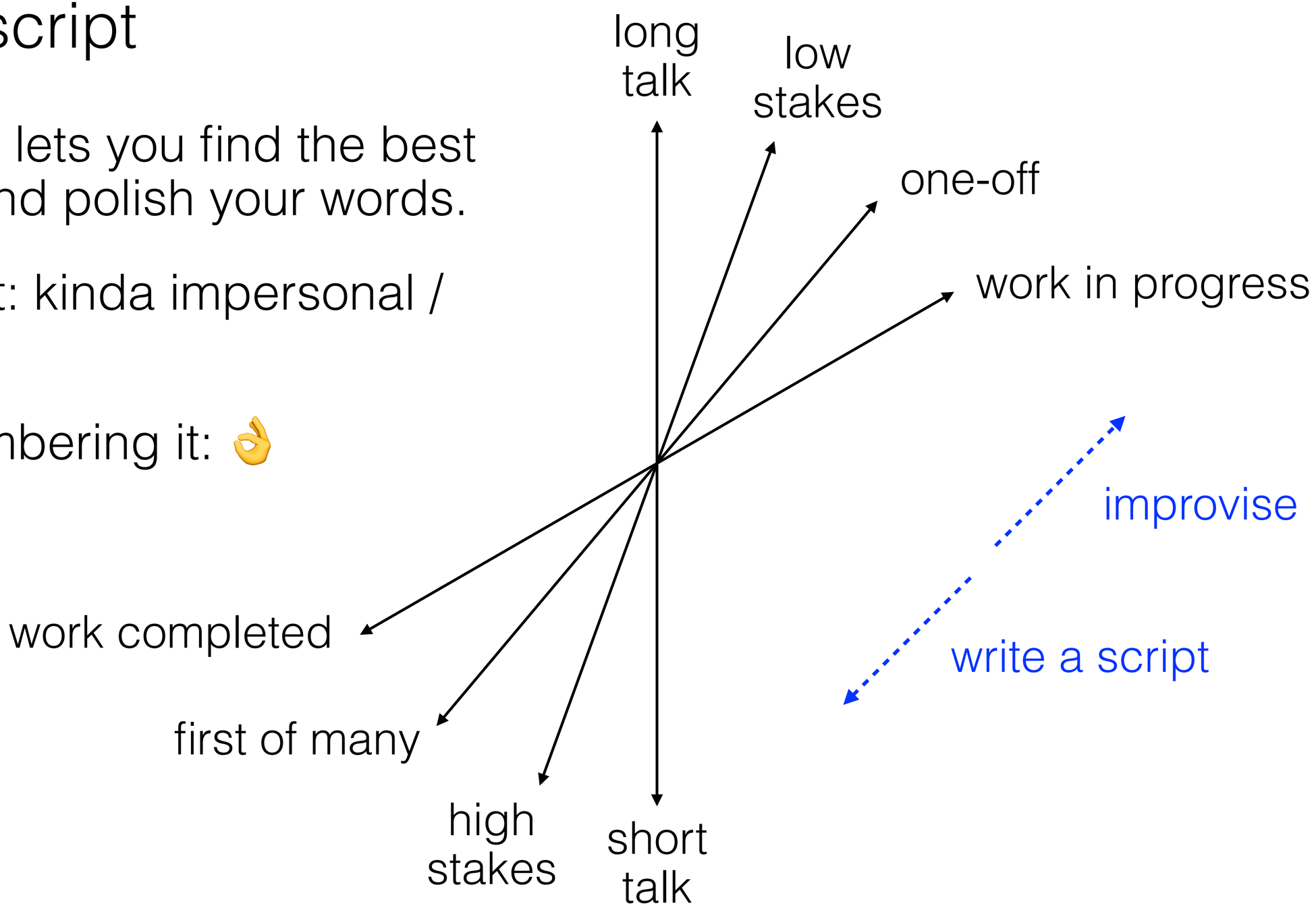
High-stakes talks: time yourself + get feedback

Consider a script

Writing a script lets you find the best explanations and polish your words.

Reading from it: kinda impersonal / robotic.

Roughly remembering it: 👌



A prompt helps you remember your script

Script:

Disease progression is stopped by successful ART.

However, delaying initiation of ART results in worse outcomes thereafter.

This was shown in the INSIGHT START randomised trial.

The trial compared outcomes in two groups: those treated only when their CD4 reached 350…

Prompt:

Disease progression

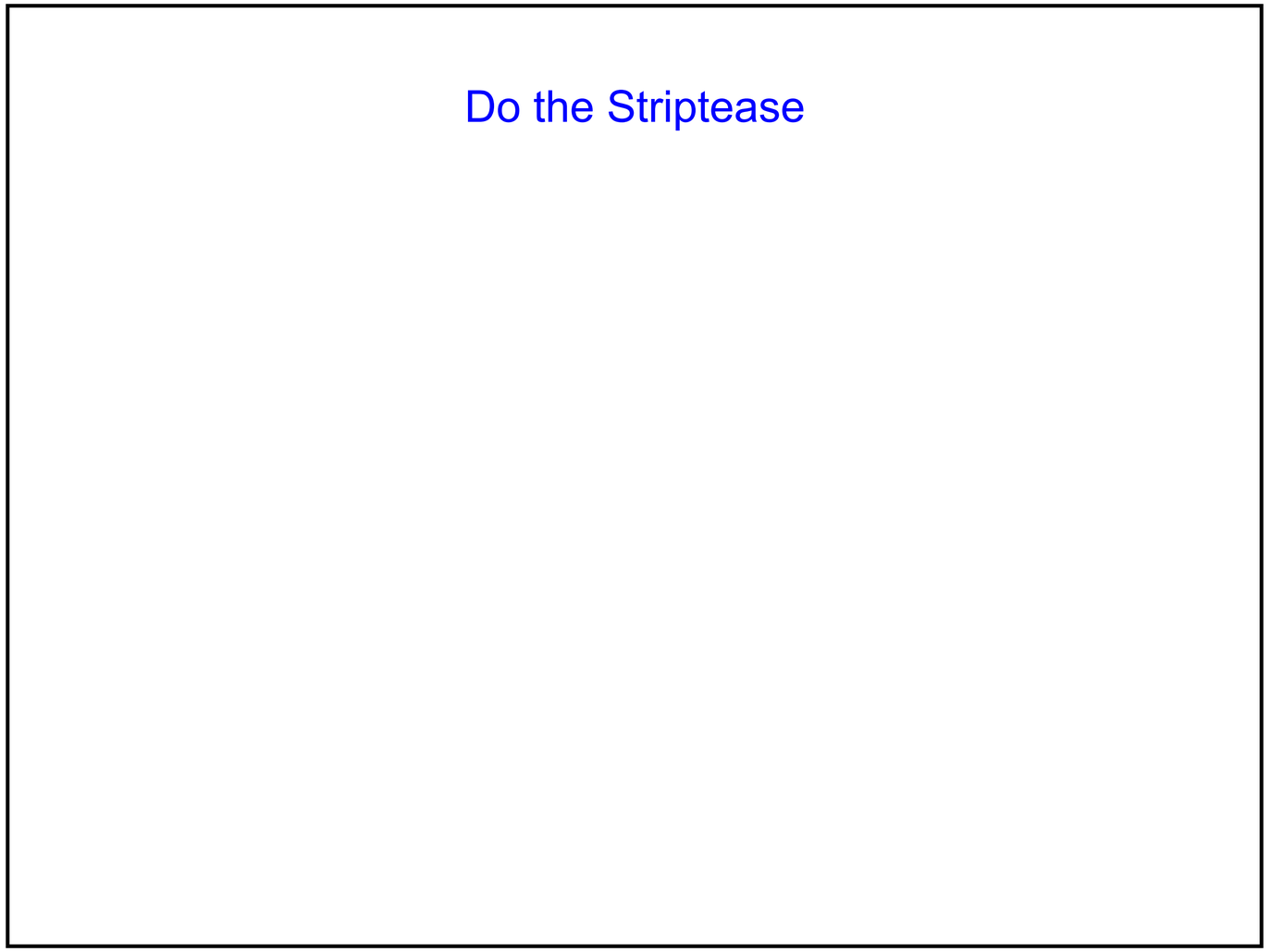
Delaying initiation

INSIGHT

Two groups

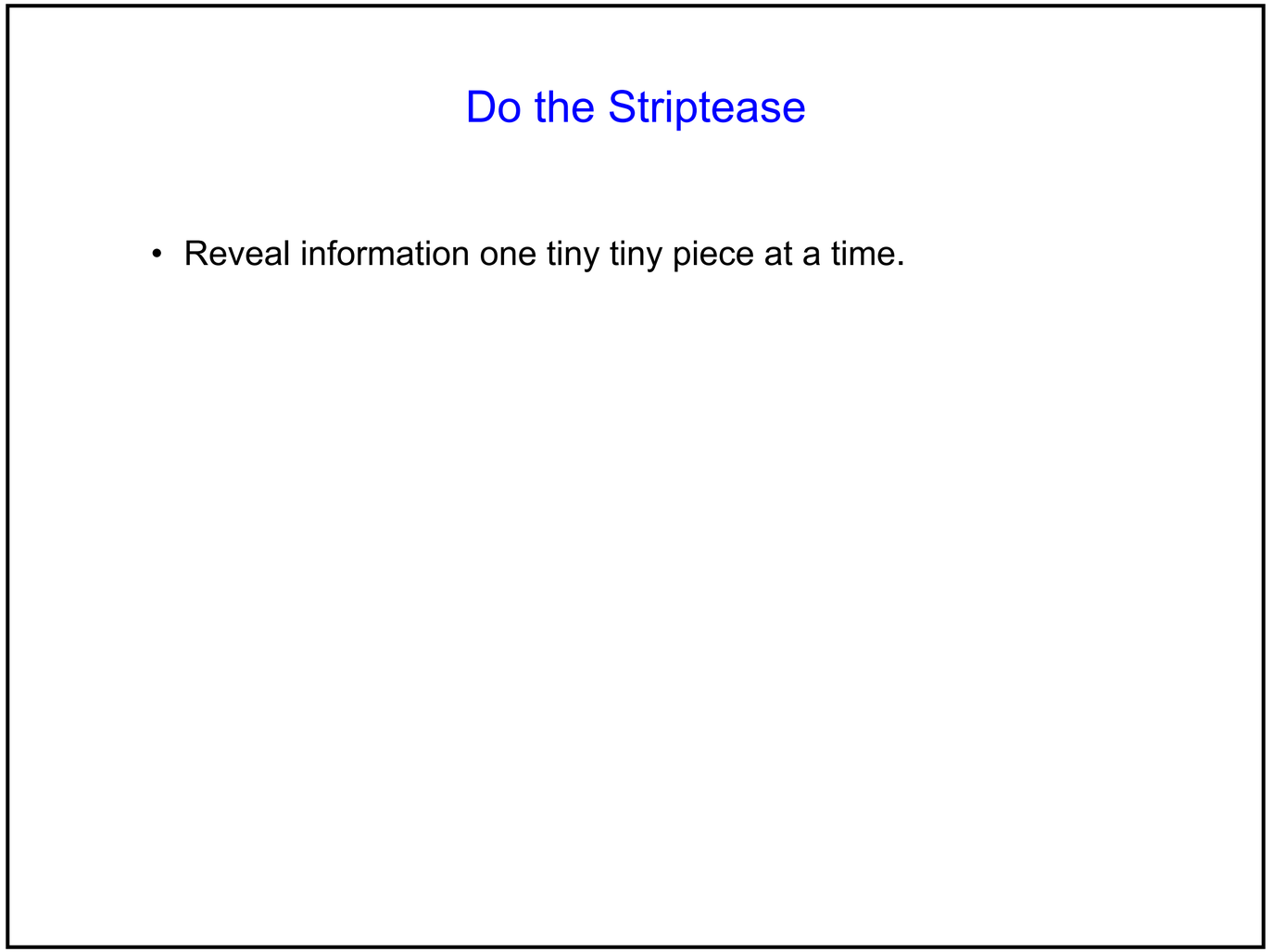
I sometimes write a prompt: extracting a few key words from each sentence in my script, so that I can remember the whole sentence just from the prompt.

Don’t be a tease



Credit: David Tong

Don’t be a tease



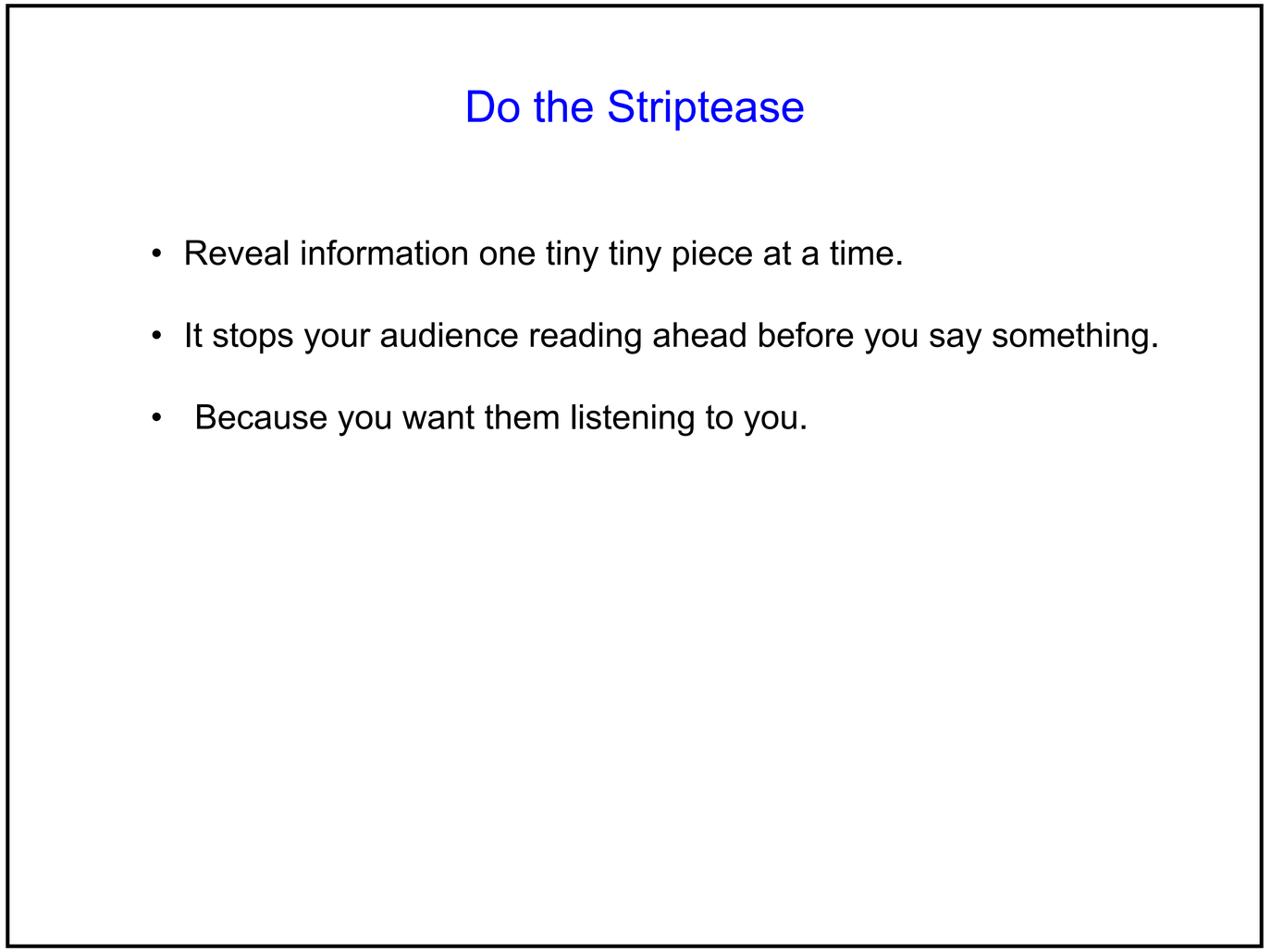
Credit: David Tong

Don’t be a tease



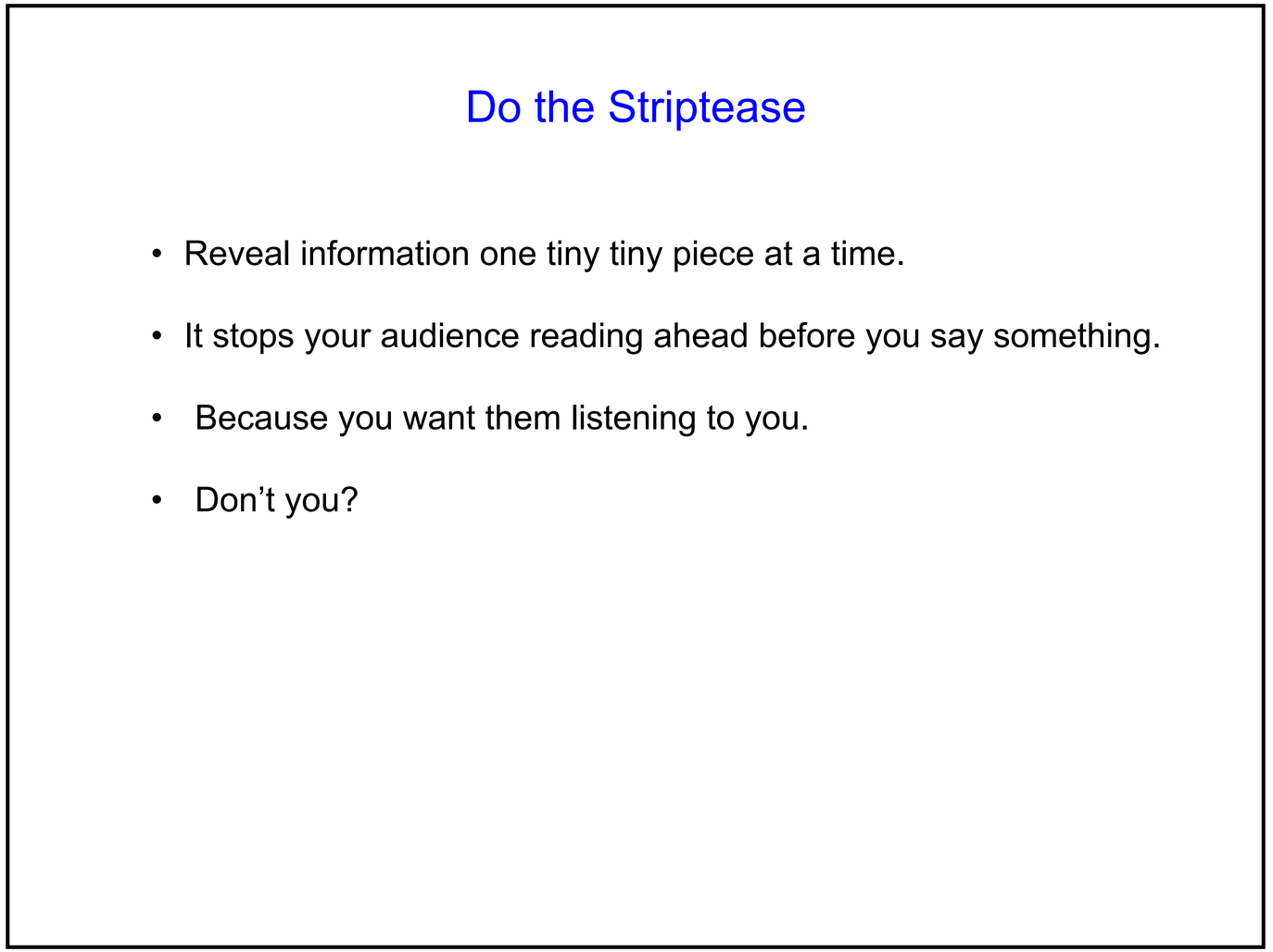
Credit: David Tong

Don’t be a tease



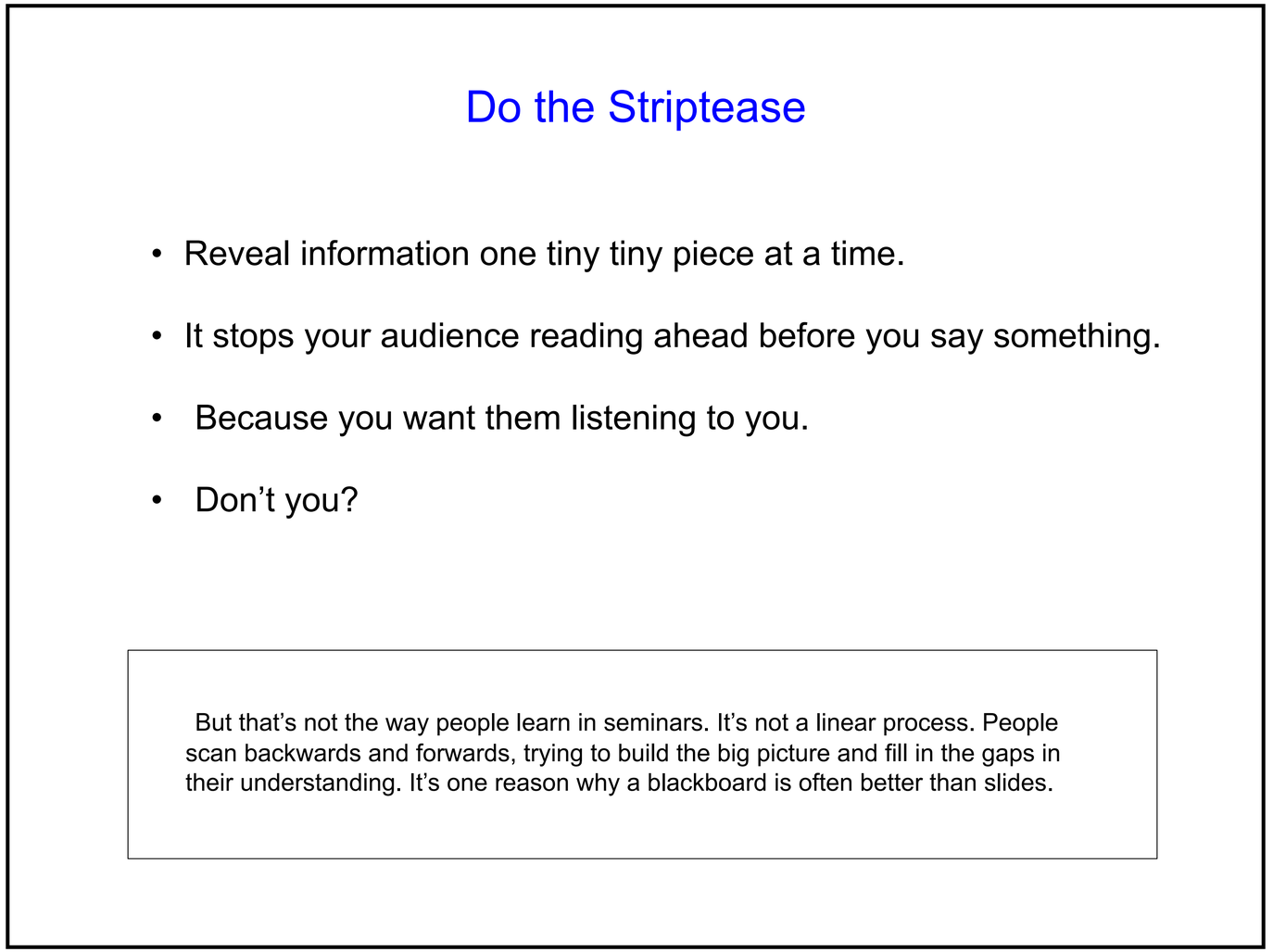
Credit: David Tong

Don’t be a tease



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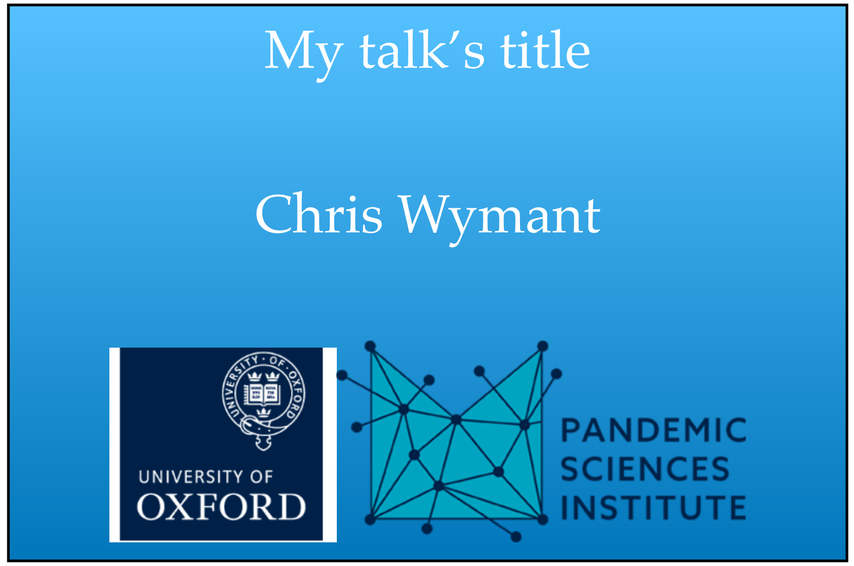
Don’t be a tease

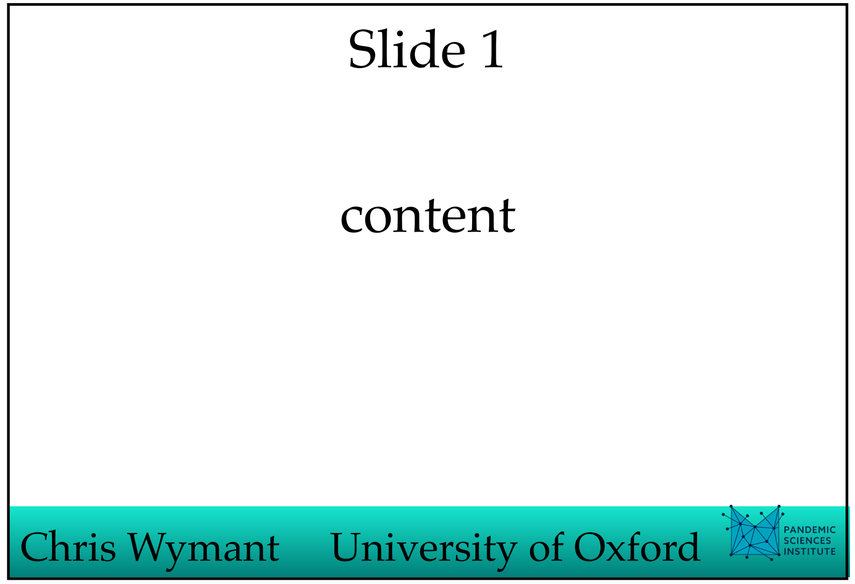


Credit: David Tong

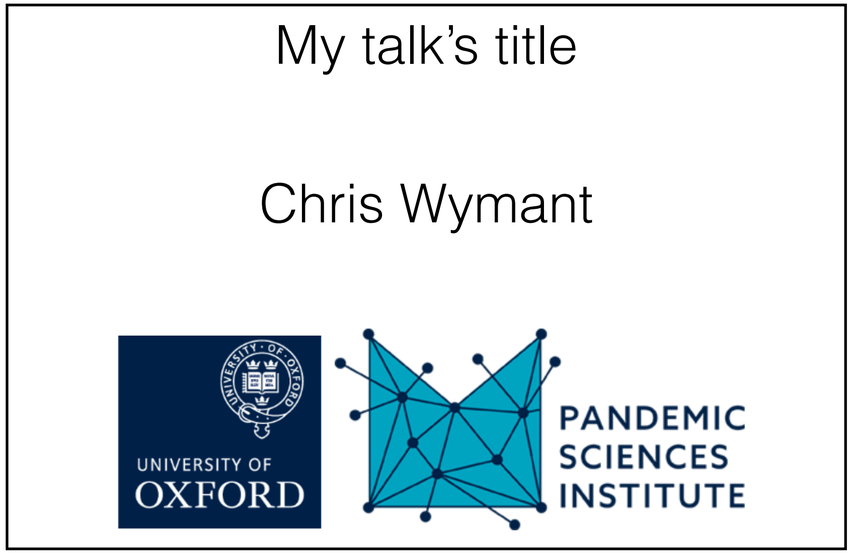
The striptease might feel attractive if you’ve put so much content on the slide that you’re worried listeners would get lost which bit you’re talking about at any given time, unless you present it to them bit by bit.
The better solution is to not put that much content on the slide.
As I said earlier about writing telegraphically: when there’s little text, listeners are able to read it quickly, so they can go backwards and forwards between text and speech as needed, which means they shouldn’t get lost which bit you’re talking about. And they can quickly read the whole thing up front to better digest the big picture of this slide before hearing you say the details, which they can’t do with a striptease.

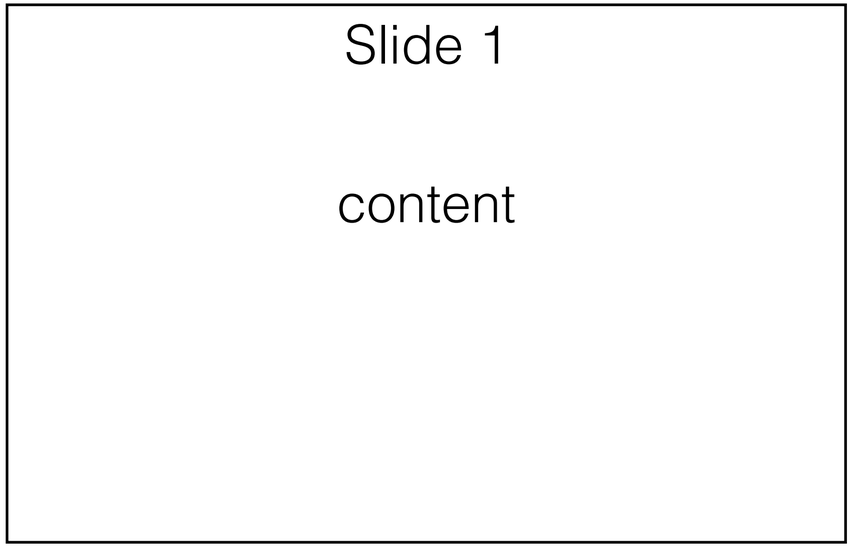
Visuals: go minimalist











Visuals: do they actually add?

Project steps:

 Collect data

 Computation

 Plotting

 Communication

or

Project steps:

Collect data

Computation

Plotting

Communication

?

I have an unpopular opinion: adding little images to each item in a list, when the text itself is already perfectly clear, just distracts my attention from the speaker briefly with no benefit. (Audience comment: it does help me, and may help for dyslexic people.)

Visuals: consider adapting plots for talks

You can re-do your own plots specifically for talks

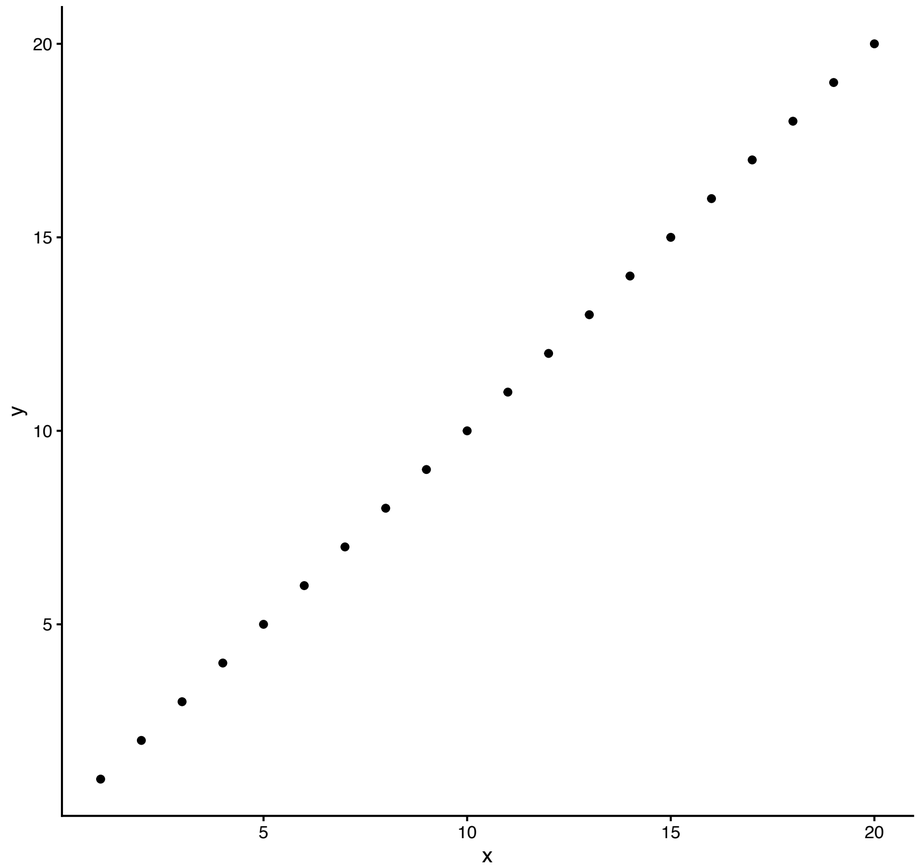
You can crop off axis labels and add clearer ones

You can add clarifications e.g. arrows + explanations

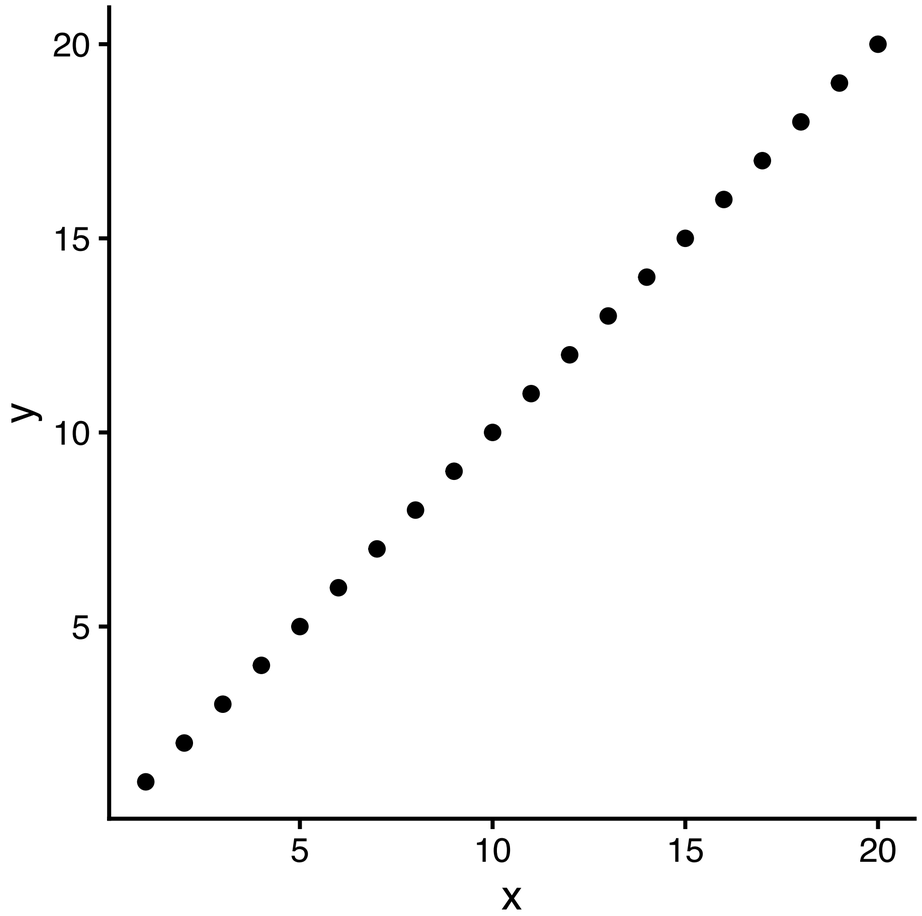
Between-plot consistency helps, e.g. colours

Visuals: plot labels are often too small

`ggsave()` default




```
ggsave(height=3.5,
       width=3.5)
```



Miscellaneous

Many citations? → footnote
[X et al., journal, year] suffices

Avoid
- excessive
  - nesting
    - of points

Limit equations, simplify where possible,
find wordy explanations.

Use titles. Consider helpful titles for lost listeners:
“The intervention worked” not “Results”.

Final slide

Your final slide stays up during questions.

Don’t waste it, e.g. “End, thank you”

Recap the main points, links / emails, …

Introductory generalities

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Logistical prep, i

Know your biological prep

> *You know what your body needs: sleep, the right amount of caffeine the right amount of time before, the right amount of food the right amount of time before…*

Plan/bring what you need
(water, clock, pointer)

Arrive early for IT setup

[pdf on the local PC] >
[plugging in your own PC] >
[fancy formats on the local PC]

Logistical prep, ii

Check all of your slides display correctly

Prevent notifications from appearing

Hybrid? Check those online can
- hear you
- see your slides in full-screen mode
- see your slides advance

Using your own computer? Close your email browser and messaging apps: popup notifications are not only distracting, they could reveal sensitive information that was intended for you only.

Introductory generalities

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Perform, engage

Giving a talk is (unfortunately?) more performative than having a conversation.
A bit of enthusiasm goes a long way.
Eye contact helps.

You're allowed to be personal.
You're allowed to be funny. But don't force it and avoid any offence.

Pleasant speaking contains some variations: pitch, emphasis, pauses etc.
Listen to professional speakers, e.g. audiobooks.

Questions

Start by inviting clarifying questions during.
Suggest other questions come afterwards.

You can interrupt a discussion with
“Let’s come back to that afterwards.”

Let the chair chair the question session.
There can be some art to it.

I think it’s best to have clarifying questions during the talk, and all other kinds of questions and comments (often of the form “have you thought about this other scenario / possible method?”) after the talk: it flows better and ensures you finish in time. Feel free to suggest that to your audience at the start, and to enforce it yourself if the chair does not. Interrupting a protracted tangential discussion with “Let’s come back to that at the end” is not rude and will be appreciated by everyone else.
During the Q&A afterwards, if your talk is being chaired by someone, allow the chair to choose who gets the next question. It’s specifically their job, and they might be paying more attention than you are to who raised their hand first, to ensuring diversity in the questions, to time remaining etc.

The Golden Rule

- Never never never go over time.
- Never.
- You might think that, given another five minutes, you can get more across. But you're wrong. No one is listening at that point. You are merely pissing people off.

Credit: David Tong

The End

Don’t leave that awkward silence hanging, where the audience don’t know if you’ve finished or are just having a deep thought.

That’s all I have to say. Thank you for listening.

Credit: David Tong

Key takeaways

Build content around 1-4 takeaways

Write telegraphically, up to ~5 points

Write legibly, speak audibly

Simple minimalist visuals

Script short high-stakes talks

Listen to talks critically

github.com/ChrisHIV/teaching